

Introduction to Contract Theory and Auctions
More about revenues & Revenue Equivalence Theorem

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

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Expected Payment once again...

- Recall that in both auction formats, FPA and SPA, or for short I and II respectively, we have that the **interim expected payment** for each individual buyer is given by

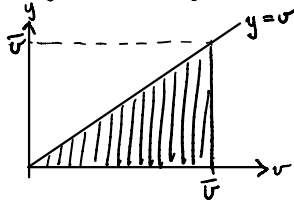
$$m^A(v) = \underbrace{G_i^{(n-1)}(v)}_{\text{Prob of Winning when value of the player is } v} \times \underbrace{E[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v]}_{\text{expected payment when winning & value of the player is } v}$$

where $A = I, II$.

- Thus, the ex-ante payment for each individual buyer is given by

Recall that $E[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v] = \frac{1}{G_i^{(n-1)}(v)} \int_0^v y g_i^{(n-1)}(y) dy$
 Thus $\hat{m}^A(v) = \int_0^v y g_i^{(n-1)}(y) dy$

The integral is over the region $\{(v, y) : 0 \leq v \leq \bar{v}, 0 \leq y \leq v\}$. Graphically



$$\begin{aligned} E[m^A(V)] &= \int_0^{\bar{v}} m^A(v) f(v) dv = \int_0^{\bar{v}} \left(\int_0^v y g_i^{(n-1)}(y) dy \right) f(v) dv \\ &= \int_0^{\bar{v}} y g_i^{(n-1)}(y) \left(\int_y^{\bar{v}} f(v) dv \right) dy \\ &= \int_0^{\bar{v}} y g_i^{(n-1)}(y) (F(\bar{v}) - F(y)) dy \\ &= \int_0^{\bar{v}} y g_i^{(n-1)}(y) (1 - F(y)) dy \end{aligned}$$

Interchanging the order of integration

Auctioneer's Revenue once again...

- Note that the auctioneer's expected revenue in each auction format, R^A , is given by

$$\begin{aligned} E[R^A] &= n E[m^A(V)] = n \int_0^{\bar{v}} y(1 - F(y)) g_1^{(n-1)}(y) dy \\ &= \int_0^{\bar{v}} y n (1 - F(y)) g_1^{(n-1)}(y) dy \\ &= E[Y_2^{(n)}] \end{aligned}$$

But $n(1-F(y))g_1^{(n-1)}(y) = g_2^{(n)}(y)$
that is, the pdf of $Y_2^{(n)}$,
i.e., the second highest order
statistic when the sample size
is n .

$\therefore$ In both the FPA & SPA, the expected revenue of the auctioneer is the same and is equal to the expectation of the second highest order statistic with a sample size of n .

Revenues from FPA vs. Revenues of SPA

- Although we know that $E[R^I] = E[R^{II}]$, it is not the case that R^I and R^{II} are the same random variable.

Example 7.1

Recall example 6.2, two bidders with $V_1, V_2 \sim U[0, 1]$ i.i.d. and $\beta^I(v) = \frac{v}{2}$ and $\beta^{II}(v) = v$. Then, by direct calculation:

Remark: the calculations in this example only apply here!

$$R^I = \max\{\beta^I(V_1), \beta^I(V_2)\} = \beta^I(\max\{V_1, V_2\}) = \beta^I(Y_1^{(2)}) = \frac{Y_1^{(2)}}{2}$$

$$R^{II} = \min\{V_1, V_2\} = Y_2^{(2)}$$

The CDF of $R^I = Y_1^{(2)}/2 \sim 2y - y^2$ if $y \in [0, 1]$

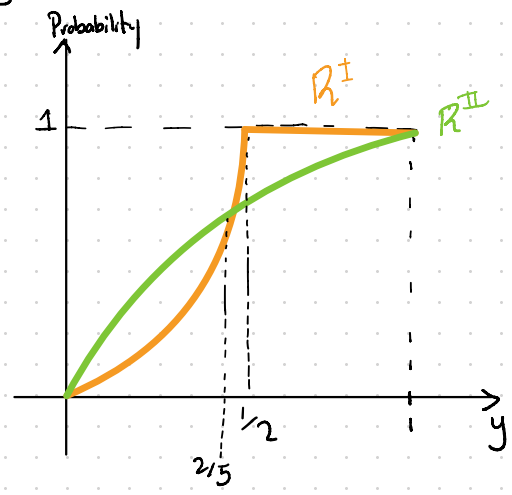
The CDF of R^I is obtained as follows
 $\Pr(R^I \leq y) = \Pr(Y_1^{(2)}/2 \leq y) = \Pr(Y_1^{(2)} \leq 2y)$
 $= (2y)^2 = 4y^2$ if $0 \leq 4y^2 \leq 1$
 $= 4y^2$ if $0 \leq y \leq \frac{1}{2}$

That is $\Pr(R^I \leq y) = \begin{cases} 0 & \text{if } y < 0 \\ 4y^2 & \text{if } 0 \leq y \leq \frac{1}{2} \\ 1 & \text{if } y \geq \frac{1}{2} \end{cases} = \min\{4y^2, 1\}$

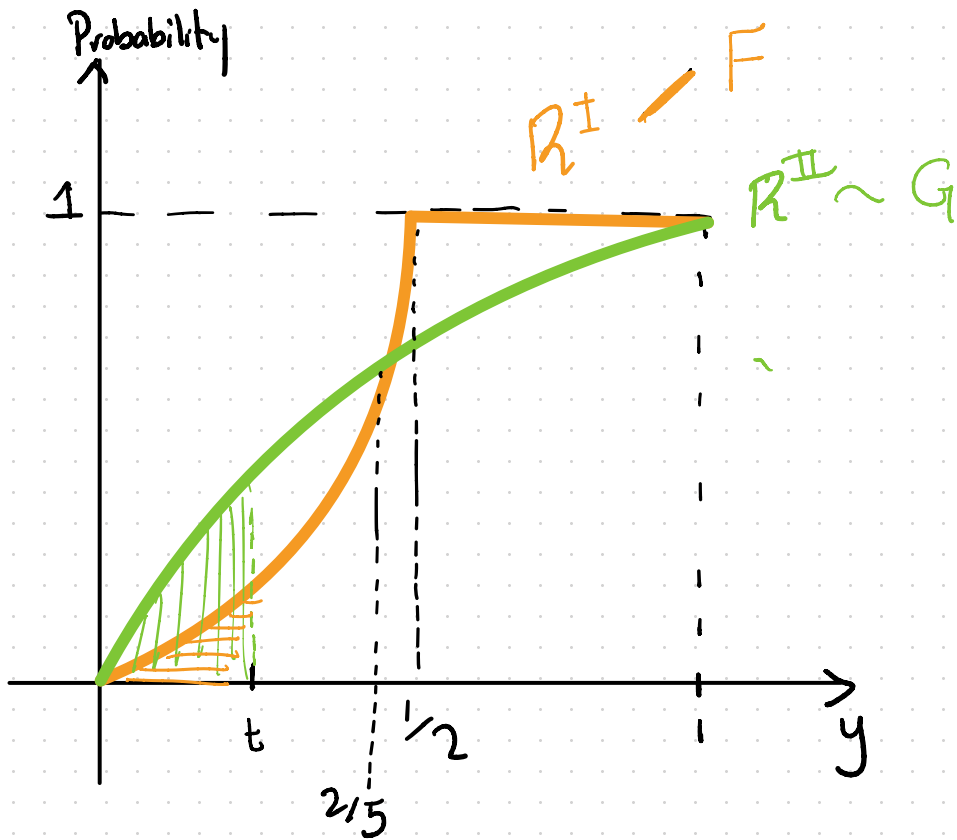
Plotting the two CDF's

$R^I \sim \min\{4y^2, 1\}$

$R^II \sim \min\{y(2-y), 1\}$



Although $E[R^I] = E[R^{II}]$, it is clear their distributions are not the same.



$$\forall t \int_0^t G(y) dy \geq \int_0^t F(y) dy$$

- Let F_I be the distribution of R^I and F_{II} be the distribution of R^{II} .
- Example 7.1 suggests that F_{II} is a **mean-preserving spread** of F_I .
- This means that from the perspective of the auctioneer, the SPA is **riskier** than the FPA.
- Indeed, any **risk-averse auctioneer** would strictly prefer a first-price auction versus a second-price auction (assuming the bidders are risk-neutral).

Proposition 7.2

In the SIPV auction model, the distribution of equilibrium payments in a second-price auction is a mean-preserving spread of the distribution of equilibrium payments in a first-price auction. That is, F_I **second-order stochastically dominates** F_{II} .

Proof.

$$R^{\text{II}} = Y_2^{(n)}, \quad R^{\text{I}} = \beta(Y_1^{(n)}), \quad \text{where} \quad \beta^{\text{I}}(x) = E[Y_1^{(n-1)} \mid Y_1^{(n-1)} \leq x]$$

$$\begin{aligned} E[R^{\text{II}} \mid R^{\text{I}} = r] &= E[Y_2^{(n)} \mid \beta(Y_1^{(n)}) = r] = E[Y_2^{(n)} \mid Y_1^{(n)} = \beta^{-1}(r)] \\ &= E[Y_1^{(n-1)} \mid Y_1^{(n-1)} < \beta^{-1}(r)] = \beta(\beta^{-1}(r)) = r \end{aligned}$$

$$\therefore E[R^{\text{II}} \mid R^{\text{I}} = r] = r$$

$$\text{Let } \varepsilon = R^{\text{II}} - E[R^{\text{II}} \mid R^{\text{I}}] \Rightarrow E[\varepsilon \mid R^{\text{I}}] = 0$$

$$\text{We can write } R^{\text{II}} = R^{\text{I}} + \varepsilon$$



The Revenue Equivalence Theorem

- Consider a SIPV sealed-bid auction described by (q, p)
 - Recall that $q : [0, \infty)^n \rightarrow \Delta(N)$, gives the winning probability in the auction as a function of the vector of bids.
 - $p : N \times [0, \infty)^n \rightarrow \mathbb{R}^n$ gives the payment vector for each player as a function of the identity of the winner and the vector of bids.
- Let $\beta : \mathcal{V} \rightarrow [0, \infty)$ be a monotonically increasing strategy.
- Denote by $m_i(v_i) = m(q, p, \beta; v_i)$ the expected payment that buyer i with private value v_i pays in an auction method (q, p) when all the buyers implement β :

$$m_i(v_i) = \int_{v_{-i} \in \mathcal{V}^{n-1}} \sum_{i_* \in N} q_{i_*}(\beta(v_1), \dots, \beta(v_n)) p_i(i_*, \beta(v_1), \dots, \beta(v_n)) dF_{-i}(v_{-i})$$

Theorem 7.3

Let β be a symmetric and monotonically increasing equilibrium in a sealed-bid auction with independent private values satisfying the following properties:

- 1 the winner of the auction is the buyer with the highest private value.
- 2 the expected payment by a buyer with private value 0 is 0.

Then

$$m_i(v_i) = G_1^{(n-1)}(v_i) \mathbb{E}[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v_i] = \int_0^{v_i} y g_1^{(n-1)}(y) dy$$

- Property (1) is called the **efficiency condition**: an efficient auction is one in which, in equilibrium, the object is allocated to the buyer who most highly values it.
- If the seller is willing to sell the object at any nonnegative price, and since the private values of the buyers are nonnegative, an efficient auction **allocates the object with probability 1**.
- What is important about theorem 7.3: the right hand side shows that the expected payment of any bidder **does not depend on the auction method!** (Caveat: assuming the SIPV assumptions and risk-neutrality of the buyers.)

Corollary 7.4

The Revenue Equivalence Theorem In a symmetric-sealed bid auction with independent private values, let β be a symmetric and monotonically increasing equilibrium satisfying the following properties (1) and (2) from theorem 7.3. Then the seller's expected revenue is

$$\pi = n \int_{\mathcal{V}} m_i(v) f(v) dv = E[Y_2^{(n)}]$$

where

$$m_i(v) = G_1^{(n-1)}(v) E[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v].$$

Uses of the RET (Thm 7.3 & Cor 7.4)

Let's consider an all-pay SIPV auction

$$q_i(b_1, \dots, b_n) = \frac{1}{|N(b)|} \quad \text{where } N(b) = \{i \in N : b_i = \max_j (b_j)\}$$

$$p_i(i_*, b_1, \dots, b_n) = b_i \quad \text{for all } i, \quad (\text{the winner } i_* \text{ doesn't matter})$$

Let's look for a symmetric, monotonically increasing bidding strategy $\beta^{all}: \mathcal{V} \rightarrow [0, \infty)$

Assume such β^{all} exists. Note that $\beta^{all}(0) = 0$.

Then conditions from the RET hold and we can apply it

In an all-pay auction the interim expected payment is

$$\left. \begin{aligned} m_i^{all}(v) &= \beta^{all}(v) \\ m_i^{all}(v) &= \int_0^v y g_i^{(n)}(y) dy \end{aligned} \right\} \Rightarrow \beta^{all}(v) = \int_0^v y g_i^{(n)}(y) dy$$

by the RET

Verifying that it is an eq.

Say all others follow $\beta^{all}(v) = \int_0^v y g_1^{(n-1)}(y) dy$ except bidder 1

say bidder 1 bids $\beta^{all}(z)$ for some $z \in \mathcal{V}$ when her value is v .

Her expected payoff is $G_1^{(n-1)}(z) \cdot v - \beta^{all}(z)$

$$= G_1^{(n-1)}(z) \cdot v - \int_0^z y g_1^{(n-1)}(y) dy$$

$$= G_1^{(n-1)}(z)(v-z) + \int_0^z G_1^{(n-1)}(y) dy$$

Follow the same steps as the proof of Thm 6.4 to deduce that $z^* = v$

Example: Third-Price auction

Suppose that $\beta^{\text{III}} = \beta$ is a symmetric, monotonically increasing eq. strategy.

- ① of RET holds } RET hold
② $\beta^{\text{III}}(0) = 0$

The [↑] expected payment of any bidder, according RET

$$m^{\text{III}}(v) = \int_0^v y g_1^{(n-1)}(y) dy$$

On the other hand, a bidder with value v wins the auction when $v > y_1^{(n-1)}$ and the price paid by such a bidder is

$$\beta(y_2^{(n-1)})$$

$$m^{\text{III}}(v) = G_1^{(n-1)}(v) \cdot E[\beta(y_2^{(n-1)}) | y_1^{(n-1)} \leq v]$$

$$g_{1,2}^{(n)}(y_1, y_2) = \begin{cases} n(n-1)f(y_1)f(y_2)F(y_2)^{n-2} & y_1 \geq y_2 \\ 0 & \text{otherwise} \end{cases}$$

↑ I need $g_{1,2}^{(n-1)}(y_1, y_2) = \frac{(n-1)(n-2)f(y_1)f(y_2)F(y_2)^{n-3}}{\text{when } y_1 \geq y_2}$

$\Pr(y_1^{(n-1)} \leq v) = G_1^{(n-1)}(v)$

PDF $y_2^{(n-1)} | y_1^{(n-1)} \leq v$

$$g_2^{(n-2)}(y_2 | Y_1^{(n-1)} \leq v) \stackrel{(*)}{=} \frac{\Pr(Y_2, Y_1^{(n-1)} \leq v)}{\Pr(Y_1^{(n-1)} \leq v)} = \frac{(n-1) \underline{f(y_2)} \underline{F(y_2)}^{n-3} (F(v) - F(y_2))}{F(v)^{n-1}}$$

$$\begin{aligned} \Pr(Y_2, Y_1^{(n-1)} \leq v) &= \int_{y_2}^v (n-1)(n-2) f(y_2) F(y_2)^{n-3} f(y_1) dy_1 \\ &= (n-1)(n-2) f(y_2) F(y_2)^{n-3} \int_{y_2}^v f(y_1) dy_1 \\ &= (n-1)(n-2) f(y_2) F(y_2)^{n-3} (F(v) - F(y_2)) \end{aligned}$$

$$g_1^{(n-2)}(z) = (n-2) f(z) F(z)^{n-3}$$

$$(*) = \frac{(n-1)(F(v) - F(y_2)) g_1^{(n-2)}(y_2)}{G_1^{(n-1)}(v)}$$

$$m^{\text{III}}(v) = G_1^{(n-1)}(v) E[\beta(Y_2^{(n-1)}) | Y_1^{(n-1)} \leq v]$$

$$= \cancel{G_1^{(n-1)}(v)} \times \frac{1}{\cancel{G_1^{(n-1)}(v)}} \int_0^v \beta(y) \cdot (n-1)(F(v) - F(y)) g_1^{(n-2)}(y) dy$$

$$\int_0^v \beta(y) \cdot (n-1)(F(v)-F(y)) g_1^{(n-2)}(y) dy = \int_0^v y g_1^{(n-1)}(y) dy$$

Differentiate wrt v both sides:

$$f(v) \int_0^v \beta(y) (n-1) g_1^{(n-2)}(y) dy = v g_1^{(n-1)}(v)$$

$$\left[\underbrace{F(v) \cdot \int_0^v \beta(y) (n-1) g_1^{(n-2)}(y) dy}_{\searrow} - \int_0^v \beta(y) (n-1) F(y) g_1^{(n-2)}(y) dy \right]'$$

$$= f(v) \int_0^v \beta(y) (n-1) g_1^{(n-2)}(y) dy + \cancel{F(v) \cdot \beta(v) (n-1) g_1^{(n-2)}(v)} - \cancel{\beta(v) (n-1) F(v) g_1^{(n-2)}(v)}$$

$$\cancel{f(v)} \int_0^v \beta(y) \cancel{(n-1)} g_1^{(n-2)}(y) dy = v g_1^{(n-1)}(v) = v \cancel{(n-1)} \cancel{f(v)} F(v)^{n-2}$$

$$\int_0^v \beta(y) g_1^{(n-2)}(y) dy = v F(v)^{n-2}$$

Diff. wrt v one more time:

$$\begin{aligned} \beta(v) g_1^{(n-2)}(v) &= F(v)^{n-2} + v(n-2) F(v)^{n-3} f(v) \\ &= G_1^{(n-2)}(v) + v g_1^{(n-2)}(v) \end{aligned}$$

$\Rightarrow$

$$\beta(v) = v + \frac{G_1^{(n-2)}(v)}{g_1^{(n-2)}(v)}$$

$$= v + \frac{F(v)^{n-2}}{(n-2) f(v) F(v)^{n-3}} = v + \frac{F(v)}{(n-2) f(v)}$$

for $\beta(v)$ to be increasing, a sufficient condition: $\frac{F}{f}$ needs to be increasing

$\frac{F}{f}$ increasing $\Leftrightarrow \ln F$ being concave $\Leftrightarrow F$ log-concave